

3. EQ_{CFG}

Let's define a new CFG $G' = (V, \Sigma, R, S)$ where S is the starting variable and every token $\sigma \in \Sigma$ has a transition rule $\sigma \rightarrow \sigma S$. If Σ includes $\{x, y\}$ then $S \rightarrow xS, S \rightarrow yS, S \rightarrow \varepsilon$, therefore $L(G') = \Sigma^*$

Now we define R as a turing machine that decides EQ_{CFG} and construct another turing machine S that decides ALL_{CFG}

S follows the rules:

"On input G where G is a CFG:

1. Construct a CFG $\langle G', \rangle$ as described above
2. Run R on G' , if R accepts then accept, otherwise reject"

In step 1. since $G' = \Sigma^*$ then R needs to decide if $G = \Sigma^*$ which is ALL_{CFG} and since we know ALL_{CFG} is undecidable then S is undecidable.